MIL-P-48166(MU)
5 March 1973

SUPERSEDING
(See 6.8)

MILITARY SPECIFICATION

PRIMER, PERCUSSION: M98

PARTS FOR AND

LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the parts for and loading, assembling and packing for one type of primer designated as Primer, Percussion: M98.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-A-2550	- Ammunition and Special Weapons, General Specification for
MIL-I-45607	- Inspection Equipment, Acquisition, Maintenance and Disposition of

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109	- Quality Assurance Terms and Definitions

FSC: 1390

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:15Z
Check the source to verify that this is the current version before use.

MIL-STD-286	- Propellants, Solid: Sampling, Examination and Testing
MIL-STD-650	- Explosive: Sampling, Examination and Testing
MIL-STD-1168	- Lot Numbering of Ammunition
MIL-STD-1235	- Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

U.S. Army Munitions Command

C-9211957	- Primer, Percussion: M98
F-9258182	- Box, Set-Up Packing Ammunition, Inner For Small Explosive Components
C-9258183	- Box, Set-UP Packing Ammunition, Outer For Small Explosive Components
C-9258184	- Box, Packing, Ammunition, For Small Explosive Components

PUBLICATIONS

ARMY

ET-9211957	- Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer).

2.2 Other publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

. Title 49	- Transportation, Parts 0-190

(The Code of Federal Regulations is available from the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402. Orders should specify, "49 CFR 0-190 (latest revision)".)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:15Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with the applicable drawings and specifications.

3.2 Primers.-The primers shall comply with all requirements specified on Drawing (Dwg.) C-9211957, all associated drawings, and with all requirements specified in applicable specifications.

3.3 Moisture content of primer mixture.-The moisture content of the primer mixture at the time of loading shall not exceed 0.30 percent when tested as specified in 4.4.1.

3.4 Functioning.-The primer shall explode under the impact of a ball weighing two (2) ounces, maximum (max.), falling a max distance of three and one-half (3.50) inches when tested as specified in 4.4.2.

3.5 Output.-The explosive output shall indent the lead disc for a depth of 0.035 inch, minimum, when tested as specified in 4.4.2.

3.6 Workmanship.-All parts and assemblies shall be fabricated, loaded and assembled in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, grease, rust, and other foreign matter. The cleaning method used shall not be injurious to any part, nor shall the parts be contaminated by the cleaning agents. Exterior surface coatings shall be continuous; however, a few light scratches not exposing base material may be permitted. All required marking and stamping shall be neat and sharply defined.

3.7 Submission of results of contractor-conducted examinations and tests.-The contractor shall forward copies of the results of any of his examinations or tests to the designated address as required by the contract. (See 6.7).

3.8 First article testing.-This specification makes provisions for first article testing. Submission of first article quantity by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unlesss otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:15Z
Check the source to verify that this is the current version before use.

contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time the completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all the quality assurance provisions specified in this specification.
b. Drawing and specification number and date, together with identification and date of changes thereto.
c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review, with the exception that a statement of Findings on the Moisture Content of each explosive shall be furnished with each lot submitted for acceptance.
d. Number of items in the lot.
e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First article inspection.-

4.2.1 Submission.-The contractor shall submit a first article quantity as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.2.2. (See 6.1) The first article sample shall consist of one-hundred and fifty (150) primers.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:15Z
Check the source to verify that this is the current version before use.

All parts and completed assemblies submitted shall have been produced by the contractor (or furnished by a supplier) using the same production processes, procedures and equipment as will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same sources of supply as will be used in regular production. The sample shall be accompanied by certificates of conformance. A first article quantity, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process (including heat treatment procedures and equipment) material used, drawing specification or source of supply as to significantly affect product uniformity as determined by the Government. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample. A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer.

4.2.2 Inspection to be performed.-All primers shall be subjected by the Government to the examination specified in 4.3.2.2. In addition, the following tests shall be performed:

4.2.2.1 Functioning (See 3.4).-One-hundred (100) primers shall be subjected to this test. The test shall be performed as specified in 4.4.2 using equipment in accordance with 4.3.4.

4.2.2.2 Output (See 3.5).-Fifty (50) primers shall be subjected to this test. The test shall be performed as specified in 4.4.2 using equipment in accordance with 4.3.4.

4.2.3 Rejection.-If any primer fails to comply with any of the applicable requirements, the first article quantity shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any primer. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article quantity or portion thereof. Until a first article quantity is accepted, the contractor is in no way authorized by the Government to resume regular production unless otherwise directed by the Contracting Officer.

4.3 Inspection provisions.-

4.3.1 Lot formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:15Z
Check the source to verify that this is the current version before use.

is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision, and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each inspection lot shall contain:

a. Parts of one lot interfix number from one manufacturer.
b. Primer mixture from one batch or blending of batches which results in homogeneity of all priming mixture used in loading the entire primer lot.
c. Lead azide from one batch or blending of batches which results in homogeneity of all lead azide used in loading the entire primer lot.
d. Lead styphnate from one batch or blending of batches, which results in homogeneity of all lead styphnate used in loading the entire primer lot.

4.3.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (See 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:15Z
Check the source to verify that this is the current version before use.

4.3.2.1 Primer (prior to loading) (See Dwg. C-9211957).-

Categories	Defects	Method of Inspection	Defect Code (See 6.2)
Critical:	None defined		
Major:	None defined		
Minor			
201.	Foreign matter (See 3.6)	Visual	01001

4.3.2.2 Primer (See Dwg. C-9211957).

Categories	Defects	Method of Inspection	Defect Code
Critical			
1.	Explosive on exterior	Visual	02001
2.	Explosive exposed (Disc punctured, cut or torn)	Visual	02002
3.	Cup cut, split or cracked	Visual	02003
Major	AQL 0.65 Percent		
101.	Weight of charge	Balance	02004
102.	Total length (prior to sealing)	Gage	02005
103.	Outside diameter, max.	Gage	02006
104.	Color improper at ends	Visual	02007
Minor	AQL 2.50 Percent		
201.	Paint on ends not dry	Manual	02008
202.	Distortion	Visual	02009
203.	Disc wrinkled or folded	Visual	02010
204.	Crimp not a full 360 degrees	Visual	02011
205.	Crack at crimp	Visual	02012
206.	Evidence of poor workmanship (See 3.6)	Visual	02013

4.3.2.3 Box, set-up, inner (after loading and prior to closing slide cover) (See Dwg. F-9258182).-

Categories	Defects	Method of Inspection	Code Number
Critical:			
1.	Cushion-filler assembly missing	Visual	03001
2.	Stiffener missing from slide cover	Visual	03002
3.	Primers improperly positioned	Visual	03003

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:15Z
Check the source to verify that this is the current version before use.

Major:	AQL 0.40 Percent		
101.	Release paper missing or improperly assembled	Visual	03005
Minor	AQL 1.00 Percent		
201.	Marking missing, incomplete or illegible	Visual	03006
202.	Evidence of poor workmanship (See 3.6)	Visual	03007

4.3.2.4 Box, set-up, inner (after closing slide cover) (See Dwg. F-9258182).

Categories	Defects	Method of Inspection	Code Number
Critical:			
1.	Slide cover assembled reversed	Visual	04001

Major: None defined

Minor: None defined

4.3.2.5 Box, set-up, outer (with inner set-up boxes assembled and prior to assembling top and sealing strip) (See Dwg. C-9258183).

Categories	Defects	Method of Inspection	Code Number
Critical:			
1.	Red surfaces on inner set-up boxes do not alternate on same side of outer set-up box	Visual	05001
2.	Black surfaces on inner set-up boxes do not alternate on same side of outer set-up box	Visual	05002
3.	All stiffeners not facing one direction	Visual	05003
Major	AQL 0.40 Percent		
101.	Box damaged to the extent that contents are exposed	Visual	05004
Minor	AQL 1.50 Percent		
201.	Marking missing, incomplete or illegible, including incorrect color code	Visual	05005

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:15Z
Check the source to verify that this is the current version before use.

202.	Evidence of poor workmanship (See 3.6)	Visual	05006
203.	Label missing or badly wrinkled	Visual	05007

4.3.2.6 Box, packing (prior to assembling cover and strapping (See Dwg. C-9258184).

Categories	Defects	Method of Inspection	Code Number
Critical:	None defined		
Major	AQL 0.40 Percent		
101.	Waterproofness of bag destroyed separation of heat seal, or rip, tear, or puncture	Visual	06001
102.	Cushioning material incomplete or missing from around outer set-up boxes	Visual	06002
Minor	AQL 1.00 Percent		
201.	Evidence of poor workmanship (See 3.6)	Visual	06003
202.	Marking misleading or unidentifiable, including incorrect color code	Visual	06004

4.3.2.7 Box, sealed wooden packing (See Dwg. C-9258184).

Categories	Defects	Method of Inspection	Code Number
Critical:	None defined		
Major	AQL 1.00 Percent		
101.	Hardware missing, broken or loose	Visual-Manual	07001
102.	Strapping missing, broken or loose	Visual-Manual	07002
103.	Contents exposed	Visual	07003
104.	Board broken or split	Visual	07004
105.	DOD symbol misleading or unidentifiable	Visual	07005
Minor	AQL 2.50 Percent		
201.	Hardware improperly engaged	Visual-Manual	07006
202.	Contents loose	Manual	07007

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:15Z
Check the source to verify that this is the current version before use.

203.	Strapping improperly assembled	Visual	07008
204.	Marking misleading or unidentifiable, including incorrect color code	Visual	07009
205.	Car seal missing, unsealed, or improperly positioned	Visual	07010

4.3.3 Testing.

4.3.3.1 Moisture content of primer mixture (See 3.3) Major defect (Code Number 08001).-At least one six (6)-gram sample of primer mixture shall be selected for the moisture content determination from that quantity to be used in each day's production of primers. Sample primer mixture shall be selected at the site of, and immediately before loading. If the moisture content exceeds the specified requirement, the primer mixture shall not be used until corrective action has been taken as verified by repeating this test. Primers already loaded with non-conforming primer mixture shall be rejected. The test shall be performed as specified in 4.4.1.

4.3.3.2 Functioning and output (See 3.4 and 3.5) Major Defect (Code Number 09001 functioning test only).-The sampling plans for this test shall be in accordance with MIL-STD-105, using Code Letter K and an AQL of 0.40 percent. If any primer fails to comply with the requirements specified in 3.4 or 3.5, it shall be classed defective. The test shall be performed as specified in 4.4.2, using equipment in accordance with 4.3.4.

4.3.3.3 Output (See 3.5) Major Defect (Code Number 10001). Primer sensitivity shall be ascertained for each lot of primers. A sample of at least 60 primers shall be selected from each lot for this test. The staircase method shall be used to ascertain the height of 50 percent functioning (h) and standard deviation (s). The quality and mixing of ingredients shall be checked if the h and s exceed the values of 1.35 and 0.50 inches, respectively. All results shall be recorded. (See 6.7)

4.3.4 Inspection equipment.-Equipment Tabulation (ET) number 9211957 and Equipment Lists (EL's) referenced thereon identify the inspection equipment required to performed the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.3. Supply and maintenance of the equipment listed on the EL's shall be in accordance with MIL-I-45607.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:15Z
Check the source to verify that this is the current version before use.

4.3.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of this specification may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey, and whenever inspection equipment drawings are needed.

4.4 Test methods and procedures.

4.4.1 Moisture content of primer mixture.-Transfer an accurately weighed portion of approximately one gram of the primer mixture to a tared aluminum weighing dish provided with a Teflon cover while both the dish and cover are on the balance pan. Place the dish and contents, with the cover removed, in an oven and dry at 60° ± 5°C for 2 hours. The cover shall be placed in a desiccator during the drying period. After drying, transfer the dish and contents quickly to a desiccator and cool to room temperature. Replace the plastic cover and weigh. Calculate the loss in weight of the dish, cover and contents to percent moisture in the sample.

4.4.2 Functioning and output.-The primers shall be mounted in the test fixture shown on the Equipment List. A weight of two ounces, maximum, shall be dropped from a height of three and one-half (3 1/2) inches, maximum, on the firing pin. Satisfactory functioning and output will be evidenced by explosion of the primers and by the size of the depression in the disc as specified in 3.4 and 3.5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging (Level C).-The Primer, Percussion: M98, shall be preserved and packaged in accordance with Dwg. F-9258182 and C-9258183.

5.2 Packing (Level C).-The Primer, Percussion, M98, preserved and packaged as specified in 5.1, shall be packed in accordance with Dwg. C-9258184.

5.3 Marking.-Marking shall be in accordance with Dwgs. F-9258182, C9258183, and C-9258184, and Code of Federal Regulations, Title 49 Parts 171-179.

5.4 Shipping.-When primers from more than one lot are shipped as a carload, each lot shall be kept separate, and the division between lots clearly indicated to prevent mixing of the lots in transit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:15Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

c. Provisions for submission of first article samples.

6.2 Inspection code numbers.-The five-digit code numbers assigned to the inspections herein are to facilitate future data collections and analysis by the Government. These code numbers are also used to correlate the characteristics cited on Equipment Lists with the inspections listed in this specification. These code numbers shall be listed on drawings of equipment designs submitted by the contractor (See 6.3.5).

6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the referenced Equipment Lists (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Equipment lists.-Equipment lists indicate the availability of inspection equipment designs by showing in the "Part Number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the "Part Number" column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Equipment list codes.-The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.

MU - Army design, mandatory for use.

6.3.3 Army designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:15Z
Check the source to verify that this is the current version before use.

contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications: Mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an equipment list references mandatory Army Designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, with the approval of the Government, design alternate inspection equipment to facilitate his operations. Such contractor-prepared designs (See 6.3.4) or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finishes, notes and operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to Specification MIL-D-1000, Category A, Form 3. Legibility and reproducibility shall permit conventional making of clearly understandable, high-contrast reproductions. Contractors shall submit three copies of final designs as a flat set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of contractor design.-All submitted designs shall contain a reference to the applicable 5-digit code number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-QA-A-S. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract. Submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished, normally, within one month after receipt.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:15Z
Check the source to verify that this is the current version before use.

6.4 Analysis of Primer Mixture Composition.-The following procedures are for use in the event that the composition of the Primer mixture is desired for information.

6.4.1 Barium nitrate.-An accurately weighed portion of approximately 0.5 gram of primer mixture shall be transferred to a tared medium-porosity, fritted glass crucible of 20 ml. capacity Three ml, of lead azide saturated distilled water shall be added at a temperature of 50 degrees C. plus or minus 2 degrees C., agitated for 1 minute, and, if necessary, lumps shall be broken up with a rubber policeman attached to a glass rod. Suction shall be applied. This procedure shall be repeated five times, making a total of six, 3-ml. extractions. The sample in the filtering crucible shall be rinsed three times with alcohol, dried in an oven maintained at 60 degrees C. plus or minus 5 degrees C. for thirty (30) minutes, cooled in a desiccator and weighed. The loss in weight shall be calculated as the percentage of barium nitrate in the sample on a dry basis as follows:

$$\text{Barium nitrate, percent} = \frac{M(A-B)}{1-.01M}$$

where:

A = weight of crucible and sample.
B = weight of the crucible and sample after extractions with lead azide saturated water.
M = moisture content of sample obtained from 4.4.1 in percent.

6.4.2 Basic lead styphnate.-The residue obtained in the determination of barium nitrate shall be extracted with six-ml. portions of saturated ammonium acetate solution at about 25 degrees C., (hot saturated ammonium acetate shall not be used as it dissolves antimony sulfide to some extent) agitating each portion for about 30 seconds and allowing each portion to remain in contact with the residue for approximately 2 minutes. The residue shall be washed with water until the filtrate coming through the crucible appears colorless. The filtrate and washings shall be quantitatively transferred to a 250 ml. calibrated volumetric flask, the beaker washed thoroughly with distilled water and diluted to the mark. Using a calibrated pipette, a 2 ml. aliquot of this solution shall be transferred to a 50 ml. calibrated volumetric flask and diluted to the mark with distilled water. A portion of this selection shall be transferred to a corex glass spectrophotometric cell having a width of approximately 1 centimeter (cm) and its absorbance density determined at a wave length of 410 millimicrons. As reference, a similar cell containing ammonium acetate solution of the same concentration as the sample solution shall be used. Since the cell which holds the basic lead styphnate is not identical with the reference cell, the absorbance density obtained shall be corrected as described above for the difference in the

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:15Z
Check the source to verify that this is the current version before use.

amount of light which the two cells scatter and absorb. To do this, both cells shall be filled with ammonium acetate solution and the absorbance density of the cell which originally contained the basic lead styphnate solution shall be measured at a wave length of 410 millimicrons. The percentage of basic lead styphnate in the sample shall be calculated on a dry basis as follows:

$$\text{Basic lead styphnate, percent} = \frac{F(A-B)}{W(1-.01M)D}$$

where:

A = optical density of basic lead styphnate solution.
B = opitcal density of ammonium acetate solution.
D = width of corex cell, cm.
M = moisture content of sample obtained from 4.4.1 in percent.
W = weight of sample.
F = factor to be determined for each instrument on the basis of a sample of lead styphnate that is suitable for use as a standard.

6.4.3 Tetracene.-The wet residue obtained in the determination of lead styphnate shall be washed three times with alcohol, dried in an oven maintained at 60 degrees C. plus or minus 5 degrees C. for 30 minutes, cooled in a desiccator and weighed. The dried residue in the crucible shall be transferred to a 125 ml. beaker with a stream of water, 25 ml. of water shall be added, and the contents of the beaker boiled on a hot plate for 5 minutes. The contents of the beaker shall be poured into the same crucible, and the residue in the crucible washed three times with boiling water, and finally with alcohol. The crucible and contents shall be dried in an oven maintained at 60 degrees C. plus or minus 5 degrees C. for 30 minutes, cooled in a desiccator and weighed. The loss in weight shall be calculated as the percentage of tetracene in the sample on a dry basis as follows:

$$\text{Tetracene, percent} = \frac{(C-D)\ 100}{W(1-.01M)}$$

where:

C = weight of crucible and residue after extraction with saturated ammonium acetate solution.
D = weight of crucible and residue after treating with hot water
W = weight of the sample.
M = moisture content of sample obtained from 4.4.1 in percent.

6.4.4 Antimony sulfide.-The weight of the residue shall be calculated from the determination of tetracene to percentage antimony sulfide in the sample on a dry basis as follows:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:15Z
Check the source to verify that this is the current version before use.

$$\text{Antimony sulfide, percent} = \frac{(D-E)}{W(1-.01M)}$$

where:

D = weight of crucible and sample after treating with hot water.
E = weight of crucible.
W = weight of sample.
M = moisture content of sample obtained in 4.4.1 in percent.

6.4.5 Lead azide.-The lead azide content in the sample on a dry basis, shall be determined by subtracting from 100 the combined percentages of barium nitrate, basic lead styphnate, tetracene and antimony sulfide.

6.4.6 Optical density of determination.-A Quartz Ultraviolet Spectrophotometer, described in the Journal of American Optical Society, Vol. 30, Page 683, (1941), was found to be satisfactory in determining the optical density of the styphnate solution.

6.5 Sensitivity (for process control).-Primer sensitivity shall be ascertained for each lot of primers. A sample of at least 60 primers shall be selected from each lot for this test. The staircase (Bruceton) method shall be used to ascertain the height of 50 percent functioning, (h) and standard deviation (s). The quality and mixing of ingredients shall be checked if the h and s exceed the values of 1.00 and 0.30 inches, respectively. (This information shall be recorded on the data card).

6.6 Check test for possible deterioration (not applicable to new production).-The following test is designed to prevent the use of primers that have deteriorated. This check test shall be performed on primers from each lot which have been stored for more than two years, or under adverse conditions for any period of time. If the specification for the ammunition item into which the primers are assembled does not require that the check test be performed, the contracting officer should insert in the contract a provision for the performance of this check test. The contracting officer should arrange to have the contractor reimbursed for any expense incurred in his performance of the check test for possible deterioration.

6.6.1 Functioning.-The sampling plan for the test shall be in accordance with Military Standard MIL-STD-105, using Code Letter J and an AQL of 0.65 percent.

6.6.2 Output.-The sampling plan for this test shall be in accordance with Military Specification Standard MIL-STD-105, using Code letter H and an AQL of 1.00 percent.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:15Z
Check the source to verify that this is the current version before use.

6.7 Submission of results of contractor-conducted examinations and tests.-Unless otherwise specified by the Contracting Officer, the contractor shall forward requested records of examinations or tests to Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-QA-A-S.

6.8 Supersession data.-This specification includes the requirements of, and supersedes, Purchase Specification AEI-170, dated 9 June 1965.

Custodian:
Army-MU

Preparing Activity:
Army-MU

Project Number: 1390-A011

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:15Z
Check the source to verify that this is the current version before use.

FOLD

Department of the Army
Picatinny Arsenal
Dover, New Jersey 07801

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY
DOD 314

OFFICIAL BUSINESS

Commander
Picatinny Arsenal
ATTN: SMUPA-QA-A
Dover, New Jersey 07801

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:15Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD, FORM 1 JAN 65 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:15Z
Check the source to verify that this is the current version before use.